

	Shell Side	Tube Side
Fluid Name	Water	Water
Total Flow	106.26 GPM	60 GPM
Inlet Temperature	140.0 °F	60.0 °F
Outlet Temperature	120.0 °F	95.0 °F
Operating Pressure	0 PSIG	0 PSIG
Pressure Drop, Allow./Calc	6.5 / 1.84 PSIG	6.5 / 0.44 PSIG
Velocity	5.62 ft/s	2.04 ft/s
Density	61.51 lb/ft3	62.16 lb/ft3
Viscosity, Mean	0.53 cp	0.95 cp
Viscosity, Wall	0.54 cp	0.91 cp
Specific Heat	1.0 Btu/lbm,°F	1.0 Btu/lbm,°F
Thermal Conductivity	0.37 Btu/ft,h,°F	0.35 Btu/ft,h,°F
Specified Fouling Factor	0.000250 hr,ft2,°F/Btu	0.000250 hr,ft2,°F/Btu

Total Heat Exchanged	1,049,499.92 Btu/h
LMTD	52.14 °F
Overall Heat Transfer Coefficient, Clean/Dirty	305.1/262.93 Btu/hr,ft2,°F
Overall Heat Transfer Coefficient, Service	259.5 Btu/hr,ft2,°F
Surface Area	81.1 ft2
Excess Surface Area	1.32 %

Construction

	Shell Side	Tube Side
Number of Passes	1	4
Design Pressure	150PSI	125PSIG
Design Temperature	375(°F)	375(°F)
Inlet Connection (Location) Type	3.0 in NPT	4.0 in NPT
Outlet Connection (Location) Type	3.0 in NPT	4.0 in NPT

Tube Material / Diameter	316 S/S / 3/4 in
No. Tubes / Gauge / Length / Pitch	104 / 20 BWG / 48 / 15/16 TRI
Shell Material	STEEL
Head Type/Material	CAST SS316
Tubesheet Material	316 S/S
Baffle Material	STEEL
Gasket Material	COMP.FIBER
Tube to Tubesheet Joint	Roller Expanded
Approvals	ASME Sect VIII Div 1 w/U stamp.